Polytwelve Update History

Transistor Labs · current version 0.9.1

Up to 0.5.1

1. Polytwelve, a 12-voice synthesizer built from the Polysix 2 Reaktor ensemble, as AU, VST3 and standalone for macOS, then VST3 for Windows and VST3/LV2 for Linux, built automatically on GitHub.
2. Preset manager with factory and user presets, favourites, previous/next and preset files.
3. Chord Memory with a chord-name display, Unison mode, glide with legato, and the arpeggiator with host sync.
4. Chorus, Phaser and Ensemble effects, pitch and mod wheels, analog drift and detune.
5. PDF user manual in every build.

0.6.0: New panel

6. A completely new, photo-realistic hardware panel: navy metal with section boxes, cream knobs, black rubber buttons with red LEDs, slide switches, pitch and mod wheels, an amber preset display and red LED displays.
7. Layout and controls:
    - Sections in signal order: VCO, VCF, VCA, EG, MG on top; WHEELS, VOICE, ARPEGGIATOR, EFFECTS, MASTER below.
    - OCTAVE, WAVEFORM, SUB OSC, VCA MODE, MG MODE, VOICE MODE and the arpeggiator RANGE and MODE are now slide switches.
    - EFFECTS has a lit button for each effect.
    - SYNC RATE and BEND RANGE are LED displays, set with arrows, dragging or the mouse wheel.
    - The stored chord is shown next to VOICE MODE.
8. Preset strip with < > SAVE BANK ★. BANK opens the browser, which now also has Delete and Step Through Favourites Only.
9. The window can be resized and keeps its shape.
10. New About window (click the logo) with version and contact details.
11. The manual is rewritten for the new panel, and this update history ships with every build.

0.7.0: Polysix model

12. New **MODEL** switch in MASTER: **CLASSIC** (the engine so far, unchanged) or **POLYSIX**, a new engine built from the Korg Polysix service manual. Presets keep the current MODEL; projects remember it.
13. The POLYSIX model:
    - SSM2044 filter model, as on the Polysix voice board, with a clean self-oscillating sine at full resonance.
    - Cutoff, keyboard tracking (up to 150%), MG speed and delay, and PWM speed ranges from the factory specs.
    - Effects modelled on the bucket-brigade effect board, mono like the original; PHASE is now the original's short swept delay with feedback.
14. The manual has a new chapter on the two engines.

0.8.0: Calibrated against a real Polysix

15. Both engines measured against 27 recordings of a real Polysix.
16. POLYSIX model reworked to match them:
    - Cutoff knob curve (almost shut at 4, closed at 2) and keyboard tracking pivot.
    - Louder self-oscillation; pulse that closes at PW 10; real pulse, PWM and sub levels.
    - MG speed curve, a third of the depth, longer MG delay; steeper envelope times.
    - Effects rebuilt with the measured speeds, depths and levels; ENSEMBLE is now the slow, deep swirl of the real unit.
17. New **CLASSIC Calibration (A/B)** submenu in the preset menu: eight switches that each bring one of these fixes (plus an optional noise floor) into the CLASSIC engine. All off by default, so CLASSIC sounds as before.
18. Manual updated.

0.8.1: macOS fix

19. Fixed: on macOS 12 (and other versions older than the build machine) the plugin could not be opened (“This Audio Unit v2 plug-in could not be opened”). The minimum macOS is 10.13 again, and every build is now checked for it.

0.9.0: One engine, Transistor Labs

20. The POLYSIX engine is removed: the original engine is the one that stays. The MODEL switch is gone from the panel.
21. The calibration fixes for the filter, resonance, pulse width, sub, MG and envelopes are now **on by default**. Effects and Noise Floor stay off (the original effects play). Switch any of them in the preset menu (BANK) under **Calibration (A/B)**; **Defaults** restores this setting, **All Off** gives the original sound.
22. Fixed: in CHORD mode a new chord cut off the previous one. Chords are now polyphonic, so a released chord rings through its release.
23. Fixed: the MG (LFO) stopped as soon as a key was let go; it now keeps modulating through the release.
24. The vendor is now **Transistor Labs** (plugin lists, installers, About window, manual). User presets move automatically to the Transistor Labs preset folder.
25. Manual updated.

0.9.1: Two filter models

26. New **VCF Model** submenu in the preset menu (BANK):

- **OLD SCHOOL**: the filter so far, unchanged (default).
- **NEW AGE**: a new 6-stage filter with saturated resonance, cutoff-dependent resonance taming and level compensation.

27. Like the calibration switches, the model is global: presets don't change it and projects remember it.

28. Manual updated.